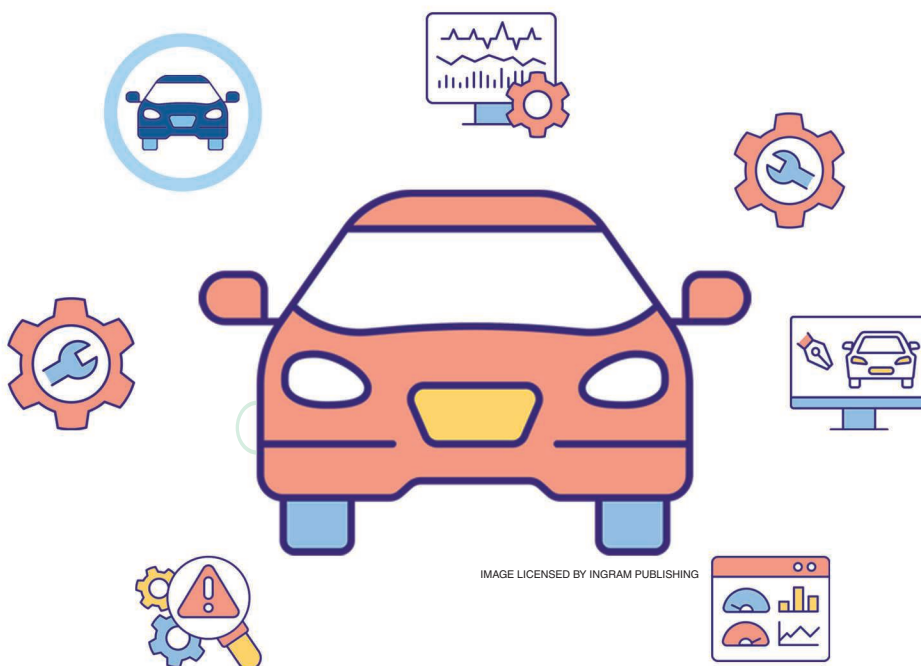


The Role of Digital Twins in Connected and Automated Vehicles



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Abstract—Digital twins found their genesis in the halls of NASA and the methods of product lifecycle management. Rapidly evolving trends around the proliferation of sensors, the Internet of Things, Industry 4.0, and cyber-physical systems have spurred the growth of digital twins. This paper reviews digital twins and their use in connected and automated vehicles (CAVs). Strictly speaking, digital twins must have communication between a physical system and its model, as opposed to similar methodologies that achieve indirect communication through iteration, or that substitute different parts of a system simulation with bits of hardware or software for testing. In practice, new methodologies for testing CAVs are sufficiently complex and difficult to apply simple labels. This is seen in our review of vehicular digital twins. Several gaps and challenges are apparent for the continued advancement of digital twin applications. We note some developing areas as traffic management centers, digital maps, onboard diagnostics, and logistics. Digital twins foster model reuse and encourage the use of multiple models at different scales of resolution. The role of digital twins will continue to grow as models become more tightly integrated to the physical systems they represent. This will drive such models towards uniqueness (matching a particular vehicle or road), adaptability (evolving with changing conditions and subject to wear and tear), and interpretability (conveying useful information to a human user). A maturing connected infrastructure and the development of smart cities will cause the number of new digital twin services to explode in a myriad of unforeseen ways.

Several trends are converging to make digital twins viable and attractive in the transportation domain, particularly in the fast-moving field of connected and automated vehicles (CAVs). The Internet of Things (IoT) movement seeks to connect large numbers of devices with embedded sensors and software [1], so large that Internet addressing will need to shift from IPv4 to IPv6 providing an address space 1,028 times larger. Developments on the IoT front promise to revolutionize manufacturing and industry. This vision is captured under the heading of Industry 4.0 [2]. The proliferation of this technology is driving an evolution in the relationships between software and hardware, between systems and models, and between complex systems and humans who need to interact with them [3]. A closely related concept is that of the cyber-physical system (CPS), in which tight integration is maintained between digital and physical components. CPS and IoT are key technologies in smart cities and intelligent transportation systems (ITS) of the future.

The idea of twinning goes back to NASA using duplicates of their Apollo spacecraft on the ground while astronauts were on mission in space. They later offered the first formal definition of a digital twin that focused on high-fidelity, lifecycle, and interdependent vehicle systems [4]. The twinning concept was specific enough to match a designated tail number. The automotive equivalent would be to match the Vehicle Identification Number (VIN) to capture all relevant information about vehicle trim, optional packages, and maintenance history.

Meanwhile, the concept of the digital twin was also defined in the context of product lifecycle management at the University of Michigan, even if the label was not yet applied [5]. At the very heart of the idea was that the digital twin

would be linked with its physical counterpart throughout its entire lifecycle so that the two parts could evolve in tandem. Grieves split the lifecycle of a product or system into four phases: *creation*, *production*, *operations*, and *disposal*. More recently, Barricelli et al. [1] defined the lifecycle phases as *designed*, *development*, *operational*, and *dismissal* [1], while Liu et al. used the terminology *design*, *manufacturing*, *service*, and *retire* [6]; same meanings—different labels. Grieves et al. furthermore asserted that traditional system engineering models such as the waterfall model, the spiral model, and the Vee model are idealistic and have the potential to lead to fragile systems. The reality, on the other hand, is that system design is highly iterative. In other words, the loop among modeling, simulation, prototyping, and testing is much tighter in practice. The digital twin concept embraces this reality and pushes it to its logical conclusion—they can in fact be simultaneous.

As computing power drove advances in simulation capabilities, modeling and simulation found new applications in research, design, testing, and validation. Small-scale applications in the 1960s gave way to well-developed simulation tools in the 1980s. Model-based design began to revolutionize manufacturing at the turn of the century, and now digital twins hold out the promise of simulation adding value throughout the entire lifecycle of a device [7]. Whereas historical modeling and simulation methodologies iterate back and forth between prototyping and building physical systems, a digital twin evolves over time with its physical counterpart throughout its life, perhaps even into its retirement. The distinctions among iteration (the typical approach in creating virtual proving grounds and tuning models), model-based design, and digital twins is diagrammed in Figure 1.

Background

The virtual proving ground was popularized in the 1990s [8] and is still in use today [9]. A virtual proving ground in the automotive space consists of a high-fidelity model of vehicle, or vehicle subsystem as well as the environment with which it interacts [10]–[12]. Ideally, an environment model would duplicate a real-world environment—a test track or proving ground, rather than a nonexistent world. Then test results obtained on the track can be used to verify and validate the models and simulation results. The third part of a virtual proving ground is the driver, whose role may be fulfilled by a model or by a human in a driving simulator [8], [13], [14]. The role of the driver has been automated with the help of standardized vehicle maneuvers and robotic steering actuators are often installed in vehicles to obtain repeatable results with J-turns, double

lane changes, understeering tests, and so on. This simplifies the problem of modeling a human driver and closes the experimental gap between virtual and real tests [15], [16].

The automotive industry was an early adopter of model-based design, specifically for electronics and embedded systems [17]. The adoption of model-based design has encouraged the use of modeling and simulation in the product design cycle and is associated with the techniques of model-in-the-loop, software-in-the-loop, and hardware-in-the-loop testing. Model-based system engineering expands the concept to systems as well as systems of systems [18], [19]. Virtual proving grounds, model-based design, and model-based system engineering have traditionally been concerned with the design, testing, verification, and validation of a product. To the extent that they result in more reliable products, they also indirectly affect later phases of

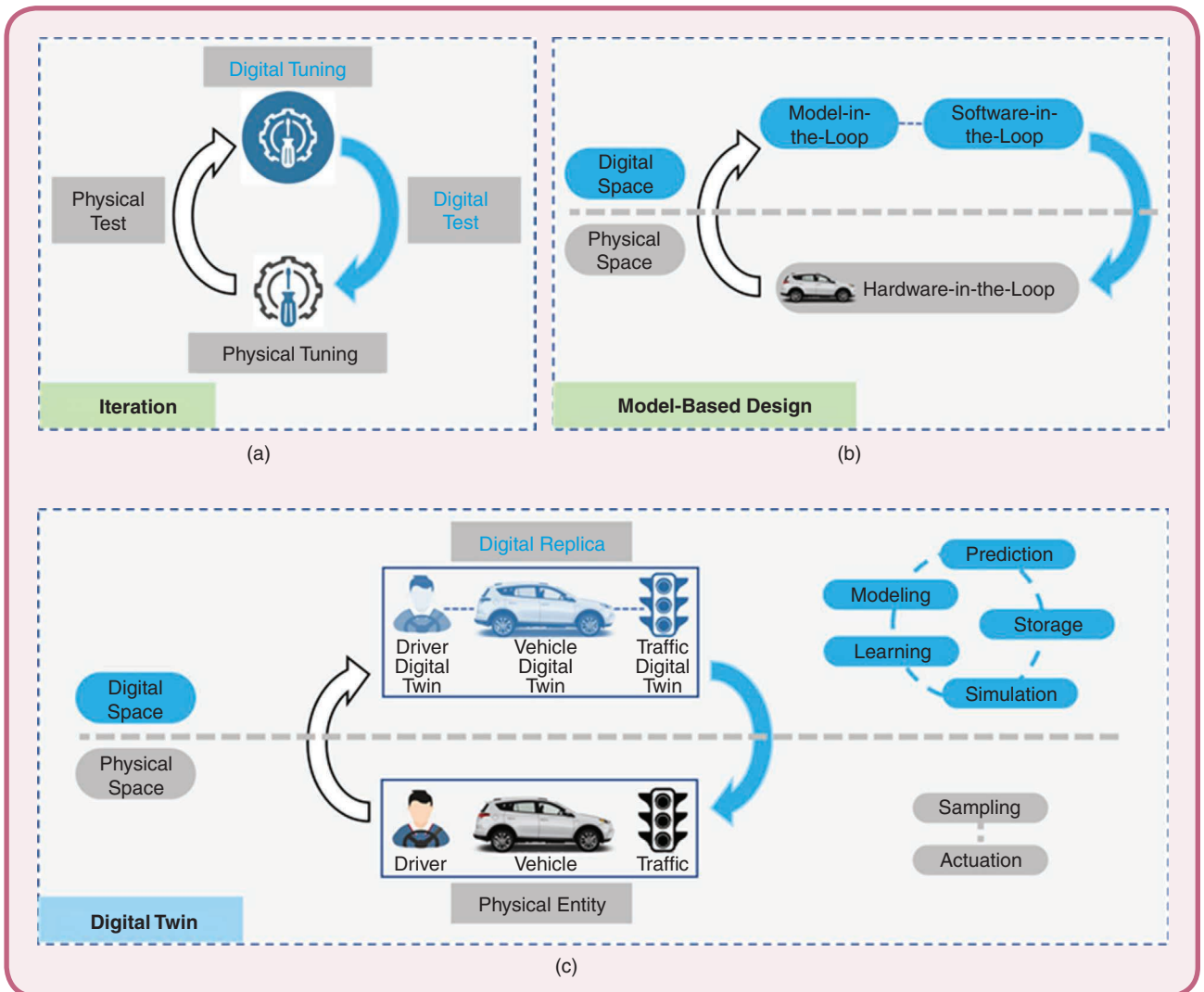


FIG 1 The simulation methodologies. (a) The iteration to convergence is commonly used to switch back and forth between the physical and digital spaces. (b) The model-based design often starts digital and incrementally swaps in physical components. (c) The digital twins maintain synchronized versions of a physical system and its digital counterpart.

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the lifecycle by reducing the occurrence of manufacturing delays and safety recalls.

Mass adoption of simulation-based methodologies requires new tools and processes geared towards the virtualization of physical components, systems, and systems of systems. Virtualization was discussed by Schoeggel et al. in the context of advanced driver assistance systems (ADAS) and automated driving systems (ADS) in 2018 [20], and a year later by Howard for electronics development and design [21]. Through the process of virtualizing and validating a vehicle, test environment, and scenarios, Shoeggel estimated that testing time could be reduced by 80 to 90%. Rather than replacing the physical with the virtual, the integration of the two into evermore complex CPS has created enormous opportunities and challenges [22]. The scope of CPS is very broad and overlaps with the definitions of other terms in this section. But in contrast to virtual proving ground, model-based design, and virtualization, CPS typically refers to an operational system rather than a design methodology. For example, the problem of distributed control of a network of CAVs, a complex CPS, was treated by Feng et al. [23].

The numbers of sensors, electronic control units, and computers have exploded over the last couple of decades. The introduction of ADAS and ADS required vehicles to become aware of their surroundings, make decisions, and act on them. This proliferation continues unabated with increasing market penetration of infotainment and active safety systems. Now the introduction of connected-vehicle technology and applications has injected an IoT ethos into the transportation system and motivated brand-new applications of modeling and simulation, including the digital twin.

What Are Digital Twins?

The presence of communication is a key differentiator for the digital twin, so much so that additional definitions for digital model and digital shadow have been introduced specifically to avoid confusion. A digital model mimics a physical object or system but there is no communication or data exchange between the two. A digital shadow has a one-way flow of communication from the physical model to the digital model. In contrast, the digital twin exchanges data

in both directions [2], [24], [25]. Fuller et al. noted several instances where digital models or shadows were misattributed as digital twins in the literature. To add to the confusion, many other terms have been used for similar applications in different fields including computational mega model, device shadow, mirrored system, product avatar, and synchronized virtual prototype, among others [1], [26].

Barricelli et al. [1] pointed out that there must be at least three types of communication around digital twins. First, the digital and physical twins must communicate; second, multiple digital twins in a complex simulated environment should be able to communicate among each other; third, there should be communication between digital twins and domain experts through user interfaces [1]. A more comprehensive breakdown is offered by the five-dimensional digital twin model, introduced by Tao [27], [28]. The five dimensions can be expressed in a formula as

$$M_{DT} = (PE, VM, Ss, DD, CN), \quad (1)$$

where PE are physical entities, VM are virtual models, Ss are services, DD stands for data, and CN denotes connections. Connections are established between each possible pairing of the other four dimensions.

The five-dimensional model adds two new ideas. The use of high-fidelity models with high-bandwidth connections to the world, and collected over long spans of time, establishes digital twins firmly in the category of big data. Even if the model is not high-fidelity, there may exist several models, each informed by a set of sensors. The model of the whole system might be constituted in the cloud and populated by multiple streams of data from the physical twin(s), generating enormous amounts of data over time. The value added by digital twins takes the form of services, the fifth dimension. Services can provide a variety of functions, such as analyzing the operation of the physical system, adapting to changes in the environment, changing its behavior, predicting failures before they occur, and more [29]–[31].

Jones et al. carried out a systematic review of digital twins and included an eight-dimensional model defined by the International Academy for Production Engineering's (CIRP) *Encyclopedia of Production Engineering* [32]. The CIRP definition mentions the specificity of a unique product, single or multiple lifecycle phases, models, information, and data. In contrast to other definitions, it does not emphasize the digital-to-physical connection, thus highlighting some of the differences in usage across different sectors and application domains.

Rasheed summarizes eight ways in which digital twins add value [26], and several of them are exemplified in the CAV digital twin applications described in this article. They include the following benefits:

- real-time remote monitoring and control
- greater efficiency and safety
- predictive maintenance and scheduling
- scenario and risk assessment
- better intra- and inter-team synergy and collaborations
- more efficient and informed decision support system
- personalization of products and services
- better documentation and communication.

Alam and Saddik proposed a digital twin architecture reference model that was designed to accommodate cloud-based communication [33]. The challenges of fusing local sensor data with cloud-based data from disparate sources were addressed using a Bayesian belief network. Further flexibility was introduced with fuzzy rule bases and hierarchical systems. The framework addresses many of the conceptual challenges of implementing a digital twin service in a cloud-based CPS. Given enough computing resources, an entire transportation system can be modeled and maintained as a digital twin in the cloud. Multiple services can then be provided to drivers, vehicles, and traffic managers based on analyses and predictions made with the digital twin.

Before presenting digital twin examples, we contrast them against the other methods described earlier. In the digital twin literature, it is tempting to apply the label to new methodologies, even if one of the more traditional terms, such as virtual proving ground, would be more accurate. While misattribution is frequent, we conjecture that the cause is simple misunderstanding or failure to agree on standard terminology. In fact, the complexity of modern design frameworks is already so great that it is difficult to tell when one methodology ends and another begins.

Nevertheless, it is useful to define terms, categorize methods, and understand where the state of the art lies. There is a tendency to equate a digital twin with a highly accurate model of a physical system or environment. In other words, as a model gets more accurate, it converges to a digital twin [34]–[36]. We take the opportunity in this article to push back against that notion and reinforce definitions of digital twins that have emerged over the last decade and were described earlier. Hallerback et al. [34] conclude with a prototype-in-the-loop method, which would meet the strict five-dimensional definition.

We note three ways that digital twins are distinct from other methodologies, which often create confusion in the literature. Communication is a key aspect of the

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three-dimensional and five-dimensional models of digital twins. In contrast, virtual proving ground and model-based design methodologies often utilize iteration between physical and virtual tests to rapidly design, validate, and test a system. While the overall time needed for such testing can be reduced dramatically, the process is still iterative and often requires manual tuning by engineers in between physical and virtual tests [37], [38]. The communication channels built into digital twins are understood to be more tightly coupled, usually simultaneous and real-time.

A second observation seems obvious but is easy to miss when reading complicated descriptions of test environments. A digital twin maintains a connection between the physical object and the model that represents it. Model-based design and model-based system engineering methodologies make use of models that work in conjunction with bits of software and hardware but are not required to pair a model with its physical counterpart. Rather, an electronic control unit is paired with a vehicle dynamics model; a real vehicle drives through a virtual environment with simulated traffic; a deep learning model for an ADS is trained using simulated scenarios [36].

A third factor that puts digital twins more in the camp of CPS than with traditional methodologies like virtual proving ground and model-based design is that digital twins are useful throughout the lifecycle of product. However, we do not see this distinction as being critical to the classification of a digital twin. One can certainly create a twin whose purpose is limited to the design or manufacturing phase of a product lifecycle as long as it has the five dimensions. As IoT and cloud-based applications mature, digital twins that had been limited in their application will naturally find new life throughout the vehicle's lifecycle. In practice, the testing framework can be so complex that different methodologies coexist or trade off responsibilities at different times [39].

Examples of Digital Twins

There are several examples in the digital twin literature geared towards the development of CAVs. It is not clear in several of these cases whether the digital twin is the correct label; however, we reiterate the fact that testing frameworks have become increasingly complex and difficult to capture with any single label. The examples documented

here are also summarized in Table 1. The included references were selected through a Google Scholar search using the terms *digital twins*, *connected and automated vehicles*, *virtual proving grounds*, *model-based design*, *cyber-physical systems*, and *parallel driving*. Search results were filtered down to focus on CAVs and exclude other digital twin applications. The cited papers invoke digital twins, either in spirit or with their language. The methodology classifications offered in Table 1 were assigned after reviewing the examples.

Transition Applications

An early application of digital twinning that sits between Industry 4.0 and ITS is instructive and has parallels to more recent applications in the CAV field. It concerns a factory floor with an unmanned autonomous vehicle. The cyber-guided vehicle described in the article is a materials-handling robot in an industrial setting [31]. The entire system was miniaturized and realized with small Arduino-powered robots. The vehicle moves around a predefined track and is tasked with delivering material to one of three different handling stations. The

digital twin was a discrete-event simulation that manages scheduling of operations in the facility. Since it was not concerned with lower-level operations such as the dynamic driving task, a simple digital representation of the vehicle sufficed.

Perhaps the most direct translation from Industry 4.0 applications to an ITS occurs in tightly controlled vehicle environments such as docks, shipping yards, and distribution centers. Barosan et al. developed a digital twin to support truck docking at a distribution center [40]. The digital twin was constructed using a model-based system engineering methodology in a modeling environment composed of IBM Rhapsody, MATLAB Simulink, and Unity. There may be several trucks driving around the dock and the simulations helped solve complex path-planning problems. The authors recognized their so-called virtual simulation environment to be a valuable testing tool for engineers, but apart from a brief mention of scaled-down trucks with Raspberry Pi controllers, little is said about the physical components. Neither did they focus on its potential to improve scheduling efficiency and driving safety during everyday operations at a functioning distribution center, a logical next

Table 1. A summary of digital twin examples related to CAVs.

Ref No.	Lead Author	Methodologies	Component(s) Twinned	Software/Frameworks	Service
[31]	E. Bottani	CPS	Environment, decision policy	C++	Industry 4.0 process development
[37]	V. Dygalo	MBD, iteration	Vehicle subsystem	MATLAB, Amesim, Carsim	ADAS design and test
[38]	H. Shikata	MBD, iteration	Vehicle, Electronic Control Unit	custom	EV charging design and test
[39]	V. Salehi	MBSE	None (all simulated)	SysML, Vector DYNA4, Cameo System Modeler, Unity, OpenDRIVE, ROS	Valet parking design and test
[47]	I. Barosan	VPG, MBSE	Vehicle (tractor-trailer)	SysML, MATLAB Simulink, Unity	Autonomous truck control design and test
[40]	J. Culley	VPG	Vehicle, sensors, track	ROS, Gazebo	Autonomous racing vehicle design and test
[41]	Y. Ge	MBD, CPS	Vehicle	not described	Autonomous vehicle design and test
[42]	Y. Feng	VPG, CPS	Vehicle	not described	CAV test
[43]	M. Szalai	VPG	Vehicle, pedestrian	Unity, SUMO	Autonomous vehicle test
[44]	G. Rong	MBD, VPG, CPS	Environment (test track)	SVL by LG, Unity, Apollo, Autware, ROS	SiL/HiL testing, V2X testing, synthetic data generation
[46]	D. J. Fremont	VPG	Environment (test track)	SCENIC, VerifAI, SVL by LG	Formal scenario-based testing
[47]	Y. Laschinsky	MBD	Vehicle	Vires VTD	ADAS design and test
[49]	X. Chen	CPS	Drivers	MQTT, Unity, MATLAB	Driver behavior risk analysis
[50]	M. Buechel	CPS	Environment, traffic	ROS, Apollo, NVIDIA DRIVE	Vehicle behavioral planning
[51]	Y. Liu	CPS	Traffic entities	Unity	ADAS
[52]	Z. Wang	CPS	Vehicle, driver, environment, traffic	not described	ADAS, performance evaluation
[53]	C. Schwarz	CPS	Vehicle subsystem, environment	MATLAB	ADAS enhanced ESC
[54]	S. Liu	CPS	Vehicle, sensors, environment	Unreal Engine	Provide optimized routes and trajectories for cooperation
[55]	F.-Y. Wang	CPS	Environment, traffic	iHorizon	ADAS, energy management

MBD: model-based design, MBSE: model-based system engineering, VPG: virtual proving ground.

step for the digital twin in the life-cycle of the facility.

An example of a transition application between model-based design and digital twin involved the design of a brake system module [37]. The authors proposed an iterative scheme in which the model was optimized to accurately represent the physical system, a process they termed *virtual and physical simulation technology*. A similar process for testing the design of an automatic electric vehicle charging system was proposed by Shikata et al. [38]. Culley et al. presented a digital twin for an autonomous racing vehicle that could swap out the real vehicle for a simulation of the vehicle, sensors, and track environment [41]. These applications share a common theme with many in the digital twin literature. The digital and physical versions of the common object do not appear to communicate in real time. Rather, the user tunes the simulation and compares its output with that of the physical system and then iterates. Model-based design commonly connects a model of a controller with the physical system, or vice versa. However, this approach and the loosely coupled communication implied by iteration do not strictly meet the definition of a digital twin. But the accurate models developed in this process could be used during operation to provide real-time diagnostics and optimize performance, thus earning the name. Alternatively, a testing paradigm like the ones described here could benefit from establishing real-time communication between the device under test and its simulation. For example, online model identification techniques could be used to keep up with changes in the hardware.

Automated Driving System Testing

The problem of comprehensively testing an ADS is so difficult and requires so much testing that it cannot be done without the aid of simulation. Towards that end, much of the recent literature is dedicated to the development of simulation frameworks that can be applied to ADS testing. A digital twin framework developed by Ge et al. proposed three levels of testing: a pure virtual test, a real sensor data-based test, and a vehicle-based test [42]. The framework is exemplary of the combined methodologies that characterize modern complex systems. Models of the vehicle, traffic, environment, sensors, and control algorithm are developed in the first phase. The second phase demonstrates a digital twin application in which real sensor data is fed into the simulation through a 4G/5G communication channel and control decisions could be fed back over the network (at the time of publication, it did not seem that the sensor data was collected in real time but fed from a data center). In the third phase, a real test vehicle is embedded

Multiple services can then be provided to drivers, vehicles, and traffic managers based on analyses and predictions made with the digital twin.

into a hybrid physical-digital environment that includes simulated traffic and pedestrians. To the extent that such testing also includes a simulation of the test vehicle, a digital twin is present. The mixture of a real vehicle with simulated traffic is something else—an evolution of a virtual proving ground into a full-fledged CPS.

This type of CPS, one that represents a hybrid between a digital twin and a virtual proving ground with a model-based design flavor, has been used with great success at Michigan's Mcity test facility [43]. Another example was developed with Unity, SUMO, and Python [44], since SUMO contains a model of the test vehicle and communicates with the physical vehicle, which meets the definition of a digital twin. Rong et al. developed a framework in Unity that integrates data from real sensors is the SVL simulator by LG [45]. The environments are labeled as digital twins because they model real-world locations, but there is no requirement in the testing paradigm that a test vehicle would need to drive on those same roads. A loose communication loop, established through iterative testing, was introduced using the SVL simulator to develop a formal testing method with the SCENIC scenario language and the VERIFAI toolkit [46]. A testing environment that uses Vires VTD software with a vehicle-in-the-loop configuration examined an advanced headlamp application called Active Safety Light [47].

Cyber-Physical Systems

One of the ironies of automation, highlighted by Bainbridge, is that as automation increases, the role of the human operator may become more challenging and one for which they are ill-equipped [48]. Another such irony is that as vehicles become more connected and automated, they need to pay more attention to the human operator, not less. A digital twin application has been proposed for CAVs that models the behavior of human drivers [49]. In fact, human behavior models constitute the paper's main digital twin, rather than vehicle or environment models. The authors recognized the social dimension to driving and the need to consider human behavior when analyzing safety risks. The novelty enabled by connected-vehicle safety systems is the sharing of driver behavioral models between vehicles. While knowledge of vehicle state (position, velocity, orientation, acceleration) can yield short-term predictions

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about future state, knowledge about driver behavior adds a layer of richness to this prediction that considers human decision-making and individual differences. The envisioned application creates a digital twin of a vehicle's operator and passes that model along to surrounding vehicles so that they can use it to make better predictions. The authors develop this idea into a risk analysis system that computes likelihood of collisions between vehicles. The idea could make automated vehicles smarter in the context of mixed traffic. The proposed digital twin is intended to function in the operational phase of the lifecycle rather than during development.

The next CAV example uses sensors around the infrastructure to monitor vehicle traffic and create its digital twin [50]. Information and recommendations derived from the traffic digital twin would then be fed to participating vehicles to improve upon their safety, mobility, and efficiency (all services in the digital twin paradigm). The authors describe the specifications and development of a prototype CAV called Fortuna. The communication channel between the vehicle and the digital twin of the environment would be accomplished through 5G cellular.

A common strategy that bridges developments in IoT, CPS, and digital twins is the adoption of cloud computing. The use of centrally managed simulations can bring to bear more computing power to model a local ITS. A novel approach for fusing data obtained from vehicle sensors to data obtained from a cloud-based digital twin was proposed to improve visual guidance information for drivers [51]. The authors lay out several challenges and algorithms for dealing with the fusion of such disparate data sources. The digital twin in this case consists of models of traffic agents including vehicles, drivers, and pedestrians, all simulated in the cloud. The authors used the Unity game engine as a simulation environment for human-in-the-loop test, but it was not a key technology for the digital twin implementation.

The idea was further developed in a collaboration between Toyota InfoTech Labs and the University of California, Riverside [52]. As before, a digital twin in the cloud simulated traffic participants and provided data to participating vehicles over a 4G-LTE cellular communication channel. The authors developed a cooperative ramp merging application as an example of what could be achieved. They found that communication delays and packet losses

could be kept to within acceptable limits and that the system could provide useful information to a driver in real time. This is an example of a communication service between the digital twin and the domain expert (driver) called out by Barricelli et al. [1].

We have described digital twins that model the driver and traffic

participants, but they must also model the driving environment. The most important piece of this environment is the road and digital maps of road networks continue to grow in fidelity and resolution to keep up with demanding applications of CAVs. Even lower-fidelity digital maps can provide useful information as a service to a connected vehicle. One way to think about digital maps is as a nontraditional type of sensor, one that is not limited by line of sight [53]. A greater challenge is to keep digital maps up to date as real-world conditions change. Recently a cloud-based digital twin environment was proposed to merge traffic information with the high definition (HD) digital map [54]. To the extent that the road digital maps can reflect real-time conditions and provide value-added services to CAVs, they represent a valuable digital twin application.

Parallel Driving

A concept very similar in scope and intent to the digital twin is called parallel driving [55] and emerged from a study of parallel control concepts in cyber-physical systems [56]. The parallel driving framework has three levels: the physical world, the mental world in which a road user develops mental models and exhibits cognitive behaviors, and the artificial world that contain computer-based models and simulations. Furthermore, parallel driving also defines services that provide value by linking the three layers in useful ways. Wang et al. provided two use cases for parallel driving. The first describes parallel testing, parallel learning and parallel reinforcement learning to combine data from the physical and artificial worlds to train a self-parking feature. The second was labeled as intelligent horizon (iHorizon) and developed an extension of eHorizon technology developed by Bosch, Continental, and HERE. They made use of digital maps and driving style recognition algorithms to make short and long-term speed predictions. The services that could be provided by such technology include traffic calming and intelligent energy management technology.

Opportunities and Challenges

Ongoing IoT and CPS trends will continue to facilitate new digital twin applications in the coming years. As CAVs gain market penetration, the benefits of an increasingly connected traffic system will begin to multiply and realize the

promise of connected safety warning systems. There are several natural lines of development for digital twins that extend already mature technologies. Other applications will be fundamentally new and shift the paradigm of what is possible.

The challenges that lie ahead for digital twins have been well documented [1], [2], [24]. Data security and privacy are omnipresent watchwords for digital twin, CPS, and IoT applications. Model interoperability and standardization enable the creation of models that can talk to one another, and its absence continues to be a barrier to large-scale systems. Issues around wealth and equity must be addressed before this (and other) technological innovations can be enjoyed worldwide. Data quality and validated models are necessary to support critical services, but verifying sufficient accuracy is no easy matter. User interfaces that allow humans to understand digital twin performance and explain communicated results are difficult to create and user trust difficult to foster. Technical limitations remain that require high-speed infrastructure and large-scale computing platforms. On the other hand, this must be balanced with the cost of developing and deploying a new technology. We mention a few opportunities specific to digital twins in the development and operation of CAVs.

Safety Critical Services

Digital twin services for CAVs range from convenience to critical. The sustainability and mobility of the transportation system will benefit from drivers receiving recommendations on lane selection, speed, and route. However, the promise of CAV technology includes safety applications that will help to drive crashes to zero. In order to field an effective safety service, the digital twin needs to be accurate, reliable, and have low latency.

Traffic Management Centers

Centrally managed locations for traffic observation and analytics will gain powerful new capabilities to dynamically advise and control traffic, from individual vehicle alerts and recommendations to adaptive control over traffic signals. This jump in capability will come from improved cloud-based communication through 5G cellular (and beyond), and from the addition of a simulation layer that creates a true digital twin of the traffic that can be used for prediction and counterfactual scenarios. The cooperative ramp merging application described earlier is only one small example of the type of service that could be supported by such a network.

Privacy and security are often presented as challenges related to increasing connectivity, and they are indeed. It will be possible to identify exact vehicles in traffic, down to their license number and VIN, more than ever before.

The novelty enabled by connected-vehicle safety systems is the sharing of driver behavioral models between vehicles.

There may need to be multiple layers of data, highly secure ones that contain personally identifiable information, and less protected ones that are scrubbed of such data. The potential fragmentation of connected transportation networks is also of concern. It may be that existing traffic management centers give up ground over time to networks privately controlled by manufacturers or other institutions. Then communication across networks to obtain complete coverage will need to be negotiated and paid for.

Digital Maps

Digital maps are in the middle of a revolution, with simple links and nodes being replaced by HD maps that contain information about lane lines, road furniture, curbs and elevation profiles, signals, and so on. It will take time for such maps to proliferate, but they offer tremendous opportunities for digital twin applications. The promise is not simply adding higher definition to the map, but to maintaining a live, dynamic digital twin of real roads. Weather, time of day, work zones, accidents, rockslides, and other events all contribute to temporary lane closures. Map status can be crowd-sourced with driver reports and using CAV sensors. Recommendations on route, lane choice, speed, and caution zones are some of the services that HD map digital twins can provide. The makers of such maps, aware of the business opportunities being created, are keeping tight control over access. While this will help keep security concerns in check, it creates other concerns around equity and accessibility to benefits provided by the technology. Standardization of formats is also required to support nation-wide services such as quickly updating work zone locations.

Onboard Diagnostics

Models created in the design phase as part of a virtual proving ground or model-based design methodology deserve a new lease on life as onboard models that operate in parallel with the vehicle. The proliferation of sensors inside the vehicle as well as outside will facilitate high-speed communication between the physical and digital twins. There are many services one could envision to make the vehicle more reliable, including real-time diagnostics and failure prediction, dynamic estimation of the coefficient of friction, more accurate service advice, and advanced controllers that optimize safety, mobility, and efficiency parameters during a drive. Additionally, the presence of onboard models creates opportunities for edge computing applications that take some of the computing load out of the cloud.

Logistics

Logistics and scheduling are digital twin applications more commonly associated with Industry 4.0 initiatives. Such applications will continue to develop in factories, shipping centers, hospitals, and airports, and they will find ways to jump into transportation and integrate with CAVs. The smart cities initiative will create channels of communication, aggregate new types of big data, and generate opportunities for digital twin services that have not yet been thought of. Road users may be more willing to plan multimodal trips that incorporate mass transit or biking as the mobility gains become clear. The challenge with this, as with all new technologies, is to ensure that the final effect makes our lives easier, better, and happier, rather than busier, more fragmented, and less private.

Conclusion

This article began as an effort to understand digital twins and the role they play in the evermore complex landscape of CAV modeling and simulation. Digital twins have enjoyed a decade of fast growth and wide adoption in various sectors including manufacturing, health care, and transportation. They can be characterized using five key dimensions: the digital model, the physical system, the communication between them, the big data that is created from this process, and the services that can be deployed to add value. Digital twins enjoy the benefits of reduced cost, reduced power consumption, and miniaturization of sensors brought about by the more fundamental trends of IoT and cloud computing.

We presented several published applications that used digital twins in the context of CAVs and ITS. It is true that some uses of the term do not fully comply with the five dimensions, and their status as authentic digital twins can be called into doubt. These cases usually corresponded to old-guard methodologies of virtual proving ground and model-based design rather than new-approaches of IoT and CPS. However, we find these distinctions to be paper-thin. There is nothing to prevent the use of five-dimensional digital twin earlier in the lifecycle of a CAV. Likewise there is no reason that models and simulation tools created during a vehicle's design cannot be deployed as a digital twin during the operational phase of its life. In some cases, it merely requires a mindset shift to reimagine a tool for development and testing into a digital twin that continues to add value in new ways.

Simulation-based testing of CAVs will continue to evolve to encompass methodologies that combine virtual proving ground, model-based design, CPS, and digital twins. As such, they defy simple classification, and the community should take care not to oversimplify them with an all-encompassing label. Digital twins foster model reuse and encourage the use of multiple models at different scales of resolution. The role of digital twins will continue to grow

as models become more tightly integrated to the physical systems they represent. This will drive such models towards uniqueness (matching a particular vehicle or road), adaptability (evolving with changing conditions and subject to wear and tear), and interpretability (conveying useful information to a human user). A maturing IoT infrastructure applied to ITS and smart cities will cause the number of new digital twin services to explode in a number of unforeseen ways.

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